Data cleaning methodology

The primary objective of this project is to construct an end-to-end data cleaning pipeline and an executive-level interactive dashboard evaluating vehicle sales transactions against established market benchmarks, specifically the **Manheim Market Report (MMR)**.

Operating from the perspective of a Pricing Analyst, the analysis seeks to uncover market realization rates, evaluate portfolio pricing efficiency, identify regional arbitrage opportunities, and model asset depreciation dynamics.

Prior to visualization, the raw dataset underwent rigorous preprocessing to address missing values, corruptions, structural shifts, and data type mismatches across more than 550,000 records.

Handling missing data

Initial data profiling revealed missing values across several key metadata attributes, starting with the Make column. Crucially, records containing missing Make entries also exhibited corresponding nulls across Model, Trim, and Body fields, totaling **10,301 rows.**

Analytical decision: Preserve revenue integrity

Because this project focuses on financial valuation and portfolio revenue, deleting these 10,301 rows would artificially deflate total sales volume and gross realized revenue. Instead of dropping these transactions, missing metadata entries were explicitly re-labeled as “Unknown Make”, “Unknown Model”, “Unknown Body”, and “Unknown Trim”.

Scalable Imputation for high-volume columns (Python vs Spreadsheet)

The Transmission column contained over **60,000 missing values**. Attempting to perform large-scale conditional imputation and string search within LibreOffice Calc caused severe memory freezing and performance latency due to dataset size.

To optimize performance, **Python (pandas)** was integrated into the workflow. Python scripts rapidly executed the imputation, replacing missing categorical entries with “Unknown” in seconds, preserving spreadsheet stability for smaller manual edits.

Detecting unquoted comma shift & file corruption

Detailed inspection uncovered structural file corruptions caused by **unquoted comma shifts** during raw data ingestion. In these corrupted rows, text entries containing internal commas shifted neighbouring fields to the right:

- 17-character Vehicle Identification Numbers (VINs) inadvertently shifted into the 2-letter State column
- Financial metrics (MMR and Selling Price) were pushed out of position or truncated into non-numeric metadata fields

Because these corrupted records destroyed the financial reliability of Selling Price and MMR, all shifted rows were systematically identified and purged from the analysis dataset.

Data standardization & value cleaning

- **Text Standardization:** Standardized state codes to uppercase (e.g., converting "ca" to "CA") to ensure proper geographical mapping
- **Scrap & Data Entry Floor Filter (<$250):** Filtered out rows where Selling Price or MMR were under $250 (frequently represented as 1-to-3-digit values). These ultra-low values indicate scrap vehicles, administrative salvage titles, or extreme entry errors that would severely distort average prices and skew percentage metrics
- **Strict Data Typing for Power BI:** Ensured numeric fields (odometer, condition, mmr, saleprice) strictly contain numbers or true blanks (null/empty strings). Inserting text strings like "Unknown" into numeric columns forces Power BI to evaluate them as text, resulting in import crashes and broken calculations